cl_modbus_tcp_relay
::CbRelayOn
+ CbRelayOn()
+ ~CbRelayOn()
+ onStateOrthogonalAllocation()
+ onEntry()
smacc2::SmaccAsyncClient
Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
smacc2::ISmaccState
+ ~ISmaccState()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition
TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
smacc2::introspection
::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
SharedPtr
vector< std::shared
_ptr< smacc2::introspection
::SmaccStateInfo > >
type_info *
map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccEventGenerator
Info > > >
std::enable_shared
_from_this< SmaccStateInfo >
map< const std::type
_info *, std::vector
< smacc2::introspection
::ClientBehaviorInfoEntry > >
map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccStateReactor
Info > > >
weak_ptr< smacc2::introspection
::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection
::SmaccStateInfo >
vector< smacc2::introspection
::SmaccTransitionInfo >
int
string
atomic< bool >
vector< std::shared
_ptr< smacc2::ISmaccClient > >
vector< std::vector
< std::shared_ptr< smacc2
::ISmaccClientBehavior > > >
mutex
#node_
+children_
+tid_
+eventGeneratorsInfo
+staticBehaviorInfo
+stateReactorsInfo
+stateMachine_
+parentState_
+transitions_
+depth_
+stateIndex_
-demangledStateName
+fullStateName
shared_ptr< rclcpp
::Logger >
vector< std::shared
_ptr< smacc2::StateReactor > >
vector< std::shared
_ptr< smacc2::SmaccEventGenerator > >
#stateInfo_
#logger_
#stateReactors_
#eventGenerators_
-isShutdownRequested_
#parentState_
-heartbeat_interval
_ms_
-port_
-slave_id_
+COIL_BASE_ADDRESS
+NUM_CHANNELS
-channel_
optional< std::future
< int > >
-onEntryThread_
-onExitThread_
smacc2::ISmaccOrthogonal
+ ~ISmaccOrthogonal()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
#clients_
-clientBehaviors_
-mutex_
-stateMachine_
-orthogonal_
-stateMachine_
#nh_
-ip_address_
-currentOrthogonal
-currentState
-mutex_
SmaccStatus
map< std::string, std
::shared_ptr< smacc2
::ISmaccOrthogonal > >
shared_ptr< smacc2
::introspection::SmaccState
MachineInfo >
recursive_mutex
atomic< StateMachineInternal
Action >
map< void *, std::shared
_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std
::pair< std::function
< std::string()>, boost
::any > >
vector< smacc2_msgs
::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
vector< smacc2::ISmaccState * >
shared_ptr< smacc2
::introspection::SmaccState
Info >
SharedPtr
#status_msg_
#orthogonals_
#stateMachineInfo_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
#timer_
#currentState_
#currentStateInfo_
#eventsLogPub_
#stateMachinePub
_
#stateMachineStatus
Pub_
#transitionHistory
Service_
#transitionLogPub_
-statusPub_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration
Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
vector< smacc2::ISmaccUpdatable * >
vector< std::vector
< smacc2::ISmaccUpdatable * > >
atomic< int64_t >
double
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
-end_
-initialized_
-rosInitialized
-signalDetector_
-smaccStateMachine_
-updatableClients_
-updatableStateElements_
-lastState_
-loop_rate_hz
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
-stateMachine_
map< smacc2::ComponentKey,
std::shared_ptr< smacc2
::ISmaccComponent > >
#stateMachine_
#components_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
#owner_
boost::signals2::signal
< Signature, optional
_last_value< typename boost
::function_traits< Signature
>::result_type >, int, std
::less< int >, function< Signature
>, typename extended_signature<
function_traits< Signature >::arity,
Signature >::function_type, boost::
signals2::mutex >
smacc2::SmaccSignal
< Signature, Combiner,
Group, GroupCompare,
SlotFunction, ExtendedSlotFunction,
Mutex >
boost::signals2::signal
< void(), optional_last
_value< typename boost::
function_traits< void()>
::result_type >, int, std
::less< int >, function< void
()>, typename extended_signature
< function_traits< void()>::arity,
void()>::function_type, boost::signals2::mutex >
boost::signals2::signal
< void(const std::string
&), optional_last_value
< typename boost::function
_traits< void(const std::string
&)>::result_type >, int, std
::less< int >, function< void(const
std::string &)>, typename extended
_signature< function_traits< void(const
std::string &)>::arity, void(const std
::string &)>::function_type, boost::signals2::mutex >
optional< rclcpp::Time >
optional< rclcpp::Duration >
boost::signals2::signal
< void(int channel, bool
state), optional_last_value
< typename boost::function
_traits< void(int channel, bool
state)>::result_type >, int, std
::less< int >, function< void(int
channel, bool state)>, typename extended
_signature< function_traits< void(int channel,
bool state)>::arity, void(int channel, bool
state)>::function_type, boost::signals2::mutex >
boost::signals2::signal
< void(int channel, const
std::string &error), optional
_last_value< typename boost::
function_traits< void(int channel,
const std::string &error)>::result
_type >, int, std::less< int >, function
< void(int channel, const std::string &error)
>, typename extended_signature< function_traits
< void(int channel, const std::string &error)>
::arity, void(int channel, const std::string &error)
>::function_type, boost::signals2::mutex >
boost::signals2::signal
< void(uint8_t states)
, optional_last_value<
typename boost::function
_traits< void(uint8_t states)
>::result_type >, int, std::
less< int >, function< void(uint8
_t states)>, typename extended_signature
< function_traits< void(uint8_t states)
>::arity, void(uint8_t states)>::function
_type, boost::signals2::mutex >
< void()>
-lastUpdate_
-periodDuration_
< void(const std::string &)>
< void(int channel,
bool state)>
< void(int channel,
const std::string &error)>
< void(uint8_t states)>
smacc2::SmaccSignal
< void()>
function< void()>
smacc2::SmaccSignal
< void(const std::string &)>
bool
modbus_t *
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ ~ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
smacc2::SmaccSignal
< void(int channel,
bool state)>
smacc2::SmaccSignal
< void(int channel,
const std::string &error)>
smacc2::SmaccSignal
< void(uint8_t states)>
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
-onFailure_
-onFinished_
-onSuccess_
-postFailureEventFn
-postFinishEventFn
-postSuccessEventFn_
+onConnectionLost_
+onConnectionRestored_
-postConnectionLostEvent
_
-postConnectionRestored
Event_
+onConnectionError_
-connect_on_init_
-connected_
-ctx_
cl_modbus_tcp_relay
::CpModbusConnection
+ CpModbusConnection()
+ ~CpModbusConnection()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ connect()
+ disconnect()
+ reconnect()
+ isConnected()
+ getContext()
+ getMutex()
+ getIpAddress()
+ getPort()
+ getSlaveId()
+ onConnectionLost()
+ onConnectionRestored()
+ onConnectionError()
# update()
- declareAndLoadParam()
-postWriteFailureEvent
_
-postWriteSuccessEvent_
+onWriteSuccess_
+onWriteFailure_
+onReadComplete_
-connectionComponent_
cl_modbus_tcp_relay
::CpModbusRelay
+ CpModbusRelay()
+ ~CpModbusRelay()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ writeCoil()
+ readCoil()
+ writeAllCoils()
+ writeAllCoils()
+ readAllCoils()
+ onWriteSuccess()
+ onWriteFailure()
+ onReadComplete()
- channelToAddress()
- isValidChannel()
-connectionComponent_
-relayComponent_
cl_modbus_tcp_relay
::ClModbusTcpRelay
+ ClModbusTcpRelay()
+ ~ClModbusTcpRelay()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ getConnectionComponent()
+ getRelayComponent()
-relayComponent_
-client_